PROJECT REPORT

ON

ENFORCING MANDATORY FIELDS USING UI POLICIES AND MIGRATING CHANGES WITH UPDATE SETS

Project Type : Naan Mudhalvan ServiceNow Project

Domain : IT Service management (ITSM)

Technology : ServiceNow Platform

Executive Summary

This project report documents the practical implementation of IT Service Management (ITSM) customizations using the ServiceNow platform, specifically focusing on data integrity and configuration management best practices. The project was undertaken as part of the "Naan Mudhalvan" skill development initiative.

The primary objective of this project was to enforce data quality within the Incident Management module by making the 'Priority' field mandatory when an incident's state moves to 'In Progress'. This requirement ensures that critical incidents are properly categorized before active work begins, thereby improving reporting accuracy and service level agreement (SLA) tracking.

The implementation involved two major technical milestones:

1. **UI Policy Configuration:** Leveraging ServiceNow's client-side logic engine to dynamically alter field behavior (mandatory status) without writing complex JavaScript code.

2. **Update Set Management:** utilizing ServiceNow's version control mechanism to capture these configurations in a portable XML format, simulating a real-world development lifecycle where changes are moved from a development environment to testing or production environments.

The project successfully demonstrated the end-to-end lifecycle of a ServiceNow customization: from requirement analysis and creating a container (Update Set) to configuration (UI Policy), verification, export, and final migration to a target instance. The successful migration was validated through rigorous testing, confirming that the business logic persisted correctly across instances.

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Chapter 1: Introduction

1.1 About Naan Mudhalvan Program

The Naan Mudhalvan program is a flagship initiative by the Government of Tamil Nadu designed to equip students with industry-relevant skills. It aims to bridge the gap between academic learning and industry requirements by providing hands-on training in emerging technologies. This project is a direct outcome of the specialized training received in Enterprise Service Management using ServiceNow.

1.2 Project Background

In modern IT organizations, the Incident Management process is critical for restoring services quickly. A common challenge faced by help desk teams is incomplete data entry. Agents often start working on tickets without setting the correct priority, leading to skewed metrics and potential SLA breaches. This project addresses this specific operational gap.

1.3 Project Objectives

- To understand the ServiceNow interface and navigation.
- To learn how to enforce business logic using UI Policies (Client-side scripting alternatives).
- To master the concept of "Update Sets" for configuration management.
- To perform a full migration cycle of customizations between instances.

1.4 Project Scope

The scope is limited to the **Incident Management** module. It covers the creation of a Local Update Set, the definition of a UI Policy for the 'Priority' field based on the 'State' field, and the export/import process of the XML file. It does not cover server-side scripting or complex workflow integrations.

1.5 Expected Outcomes

Upon completion, the system will automatically force the user to select a Priority value whenever the Incident State is changed to "In Progress". Furthermore, a reusable XML file containing this configuration will be generated.

Chapter 2: ServiceNow Platform Overview

2.1 Introduction to ServiceNow

ServiceNow is a cloud-based platform that provides software as a service (SaaS) for technical management support. It specializes in IT service management (ITSM), IT operations management (ITOM), and IT business management (ITBM). It allows users to manage projects, teams, and customer interactions via a variety of apps and plugins.

2.2 ITSM Fundamentals

IT Service Management (ITSM) refers to the entirety of activities â€“ directed by policies, organized and structured in processes and supporting procedures â€“ that are performed by an organization to design, plan, deliver, operate and control information technology services offered to customers.

2.3 Incident Management Module

Incident Management is the process responsible for managing the lifecycle of all incidents. The primary objective of Incident Management is to return the IT service to users as quickly as possible. In this project, we modify the behavior of the Incident form.

2.4 Configuration vs. Customization

Configuration involves using the native tools (like UI Policies, Business Rules) to change the system's behavior. **Customization** involves changing the core code or schema. This project focuses on Configuration, which is the preferred method for maintainability.

Chapter 3: UI Policies in ServiceNow

3.1 What are UI Policies?

UI Policies are client-side logic that offer an alternative to client scripts. They are used to dynamically change the behavior of information on a form and control custom process flows for tasks. UI policies allow you to set fields as mandatory, read-only, or visible based on certain conditions.

3.2 Use Cases and Benefits

- **No Code/Low Code:** Requires little to no scripting knowledge.
- **Performance:** Generally faster than client scripts.
- **User Experience:** Immediate feedback to the user without server round-trips.

> **Note:** UI Policies execute after Client Scripts. If there are conflicting logic, the UI Policy will generally take precedence regarding field attributes like visibility and mandatory status.

3.3 Components (UI Policy Actions)

The core of a UI Policy is the **UI Policy Action**. While the main policy defines "When" (the condition), the Action defines "What" (the effect). Actions can control:

- **Mandatory:** True/False/Leave Alone
- **Visible:** True/False/Leave Alone
- **Read-only:** True/False/Leave Alone

3.4 Best Practices

It is best practice to use UI Policies over Client Scripts for simple field attribute changes (Visible, Mandatory, Read-Only) because they are easier to debug, maintain, and upgrade.

Chapter 4: Update Sets in ServiceNow

4.1 Purpose of Update Sets

An Update Set is a group of configuration changes that can be moved from one instance to another. This feature allows administrators to group a series of changes into a named set and then move them as a unit to other systems for testing or deployment.

4.2 Update Set Lifecycle

The standard lifecycle involves:

1. **Creation:** Creating a new set and making it "Current".
2. **Capture:** Performing configurations; the system automatically records changes to the current set.
3. **Completion:** Marking the set as "Complete" once development is finished.
4. **Export:** Exporting to XML.
5. **Import:** Uploading to the target instance.
6. **Preview & Commit:** Checking for collisions and applying changes.

4.3 Migration Process

Migration ensures that development work (done in Dev) is safely promoted to Test and Production environments without manually re-doing work, reducing human error.

Chapter 5: Project Implementation - Milestone 1 (Update Sets)

This section details the practical steps taken to initiate the project by creating a container for our changes.

Activity 1: Create Update Set

Objective: To create a local update set to capture the UI Policy configurations.

Step-by-Step Procedure:

1. Open the ServiceNow instance and log in as an administrator.
2. Click on the **All** menu in the navigation bar.
3. Type "Update Sets" in the filter navigator.
4. Select **Local Update Sets** under System Update Sets.
5. Click on the **New** button in the top right corner.
6. Fill in the details:
   - **Name:** Ui policy for mandatory priority
   - **State:** In Progress
   - Description: Captures logic to make priority mandatory.
7. Click **Submit and Make Current**.

Verification: Ensure the "Up **Verification:** Ensure the "Update Set Picker" in the top banner (if enabled) shows "Ui policy for mandatory priority".

Chapter 6: Project Implementation - Milestone 2 (UI Policy)

Activity 1: Create UI Policy

Objective: Configure the logic to enforce the mandatory field.

1. Navigate to **System UI > UI Policies**.
2. Click **New**.
3. Configure the form:
   - **Table:** Incident [incident]
   - **Short Description:** Make Priority Mandatory when In Progress
   - **Conditions:** State | is | In Progress
4. Click **Save** (Right-click header > Save) to stay on the form.
5. Scroll down to the **UI Policy Actions** related list.
6. Click **New**.
7. Set the following:
   - **Field name:** Priority
   - **Mandatory:** True
   - **Visible:** Leave alone
   - **Read only:** Leave alone
8. Click **Submit**.

Activity 2: Check Update Set

Objective: Verify that the changes were captured.

1. Navigate to **System Update Sets > Local Update Sets**.

2. Open "Ui policy for mandatory priority".

3. Look at the **Customer Updates** related list.

4. Verify two records exist: one for the *sys_ui_policy* and one for *sys_ui_policy_action*.

Activity 3: Export and Import Update Set

Objective: Migrate the changes.

1. In the Update Set record, change **State** to **Complete**.
2. Click **Save**.
3. Under "Related Links", click **Export to XML**.
4. Save the downloaded XML file locally.
5. **(Simulation)** Open a different instance (or PDI).
6. Navigate to **System Update Sets > Retrieved Update Sets**.
7. Click the link **Import Update Set from XML**.
8. Upload the file downloaded in step 4.
9. Open the uploaded record (State will be 'Loaded').
10. Click **Preview Update Set**.
11. If no errors, click **Commit Update Set**.

Chapter 7: Project Implementation - Milestone 3 (Testing)

Activity 1: Testing Incident

Objective: Validate the functionality on the target instance.

Procedure:

1. Navigate to **Incident > Create New** (or open existing).
2. Ensure the **State** is "New". Notice the Priority field (standard behavior).
3. Change the **State** field to **In Progress**.
4. Observe the **Priority** field.

Expected vs Actual Results

Test Case ID	Scenario	Expected Result	Actual Result	Status
TC001	State changed to In Progress	Priority field gets a red asterisk (Mandatory)	Priority field became mandatory	PASS
TC002	Save form with empty Priority (In Progress)	System blocks save with error message	System blocked save	PASS
TC003	State changed back to New	Priority field is no longer mandatory	Mandatory flag removed	PASS

Chapter 8: Technical Specifications

8.1 System Requirements

- **Platform:** ServiceNow (Utah/Vancouver/Washington DC release)
- **Browser:** Google Chrome / Microsoft Edge (Latest Version)
- **Access:** System Administrator (admin) role

8.2 Configuration Details

The following table summarizes the technical configuration created in this project:

Configuration Type	Name	Table	Details
Update Set	Ui policy for mandatory priority	sys_update_set	State: Complete
UI Policy	Make Priority Mandatory	incident	Condition: state=2 (In Progress)
UI Policy Action	Priority Action	incident	Field: priority, Mandatory: true

Chapter 9: Challenges and Solutions

9.1 Challenges Faced

1. Update Set Selection: Initially, I forgot to switch the Update Set to "Current". This resulted in changes being captured in the "Default" update set.

2. Condition Logic: Confusion regarding the State values (backend integer values vs display labels).

9.2 Solutions Applied

1. I learned to verify the Update Set picker in the top right corner before starting any configuration. For the erroneously captured updates, I learned they can be manually moved, but it is safer to restart the activity in the correct set.

2. I used the condition builder UI which abstracts the integer values, ensuring the logic "State is In Progress" was selected correctly.

9.3 Lessons Learned

- Always check the active Update Set before working.
- UI Policies are superior to Client Scripts for simple validations.
- Previewing an Update Set is a crucial step to avoid breaking the production environment.

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Chapter 10: Project Benefits and Impact

10.1 Business Benefits

- **Improved Data Quality:** Ensures that every ticket being worked on has a valid priority assigned.
- **Better Reporting:** Management can trust the priority reports for incidents in progress.
- **Standardization:** Enforces a standard process workflow across all support agents.

10.2 Technical Benefits

- **Zero Code Maintenance:** The solution uses OOTB (Out of the Box) configuration, making upgrades seamless.
- **Portability:** The Update Set allows this feature to be deployed to any instance immediately.

Chapter 11: Future Enhancements

While the current project meets the core requirements, the following enhancements could add further value:

11.1 Dependent Fields

We could extend the UI Policy to make 'Assignment Group' mandatory as well when the state moves to In Progress, ensuring the ticket is owned by a specific team.

11.2 UI Policy Scripts

For more complex scenarios, we could utilize the "Run Scripts" option within the UI Policy to display a pop-up alert explaining *why* the priority is being mandated.

11.3 Data Policies

To ensure this rule applies even when data is imported via API or Excel (server-side), we could convert this UI Policy into a Data Policy.

Chapter 12: Conclusion

The "Naan Mudhalvan" project on ServiceNow administration provided a comprehensive hands-on experience in managing IT service workflows. By implementing a mandatory field requirement using UI Policies and successfully migrating it using Update Sets, I have gained practical skills in:

- Navigating the ServiceNow ecosystem.
- Configuring form behaviors without coding.
- Managing the software development lifecycle (SDLC) within the platform via Update Sets.

This project serves as a foundational step toward becoming a certified ServiceNow Administrator. The methodology applied hereâ€”Develop, Capture, Verify, Migrateâ€”is the industry standard for all ServiceNow implementations.

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Appendices

Appendix A: Glossary of Terms

- **ITSM:** Information Technology Service Management.
- **SaaS:** Software as a Service.
- **XML:** Extensible Markup Language, used for update set exports.
- **OOTB:** Out of the Box (standard features).
- **PDI:** Personal Developer Instance.

Appendix B: References

- ServiceNow Product Documentation (docs.servicenow.com)
- Naan Mudhalvan Learning Portal
- ServiceNow Developer Program

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